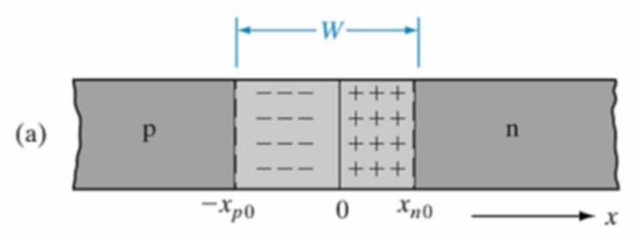
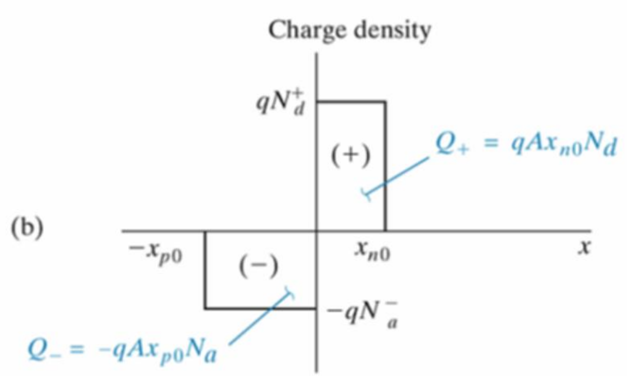
## P-N Junction

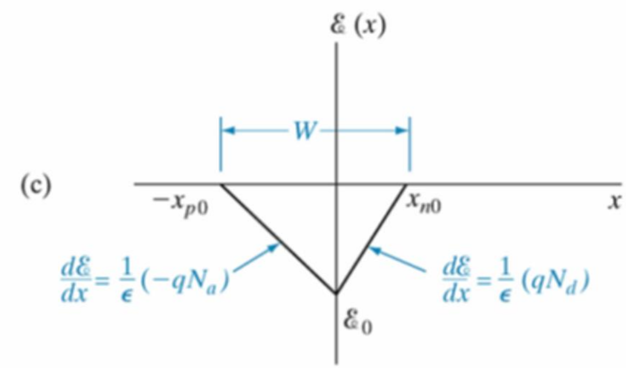
- Calculation of built-in potential ($V_{bi}$) from formula and from Energy Diagram

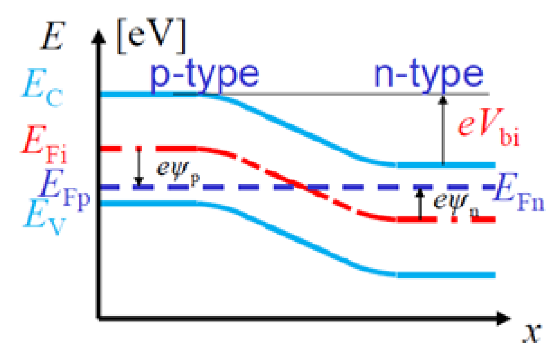
- Must know how to draw charge distribution, electric field, energy diagram at different applied biases

- Calculation of amount of charge in depletion region, maximum electric filed, total depletion width and depletion width at each side of the junction.



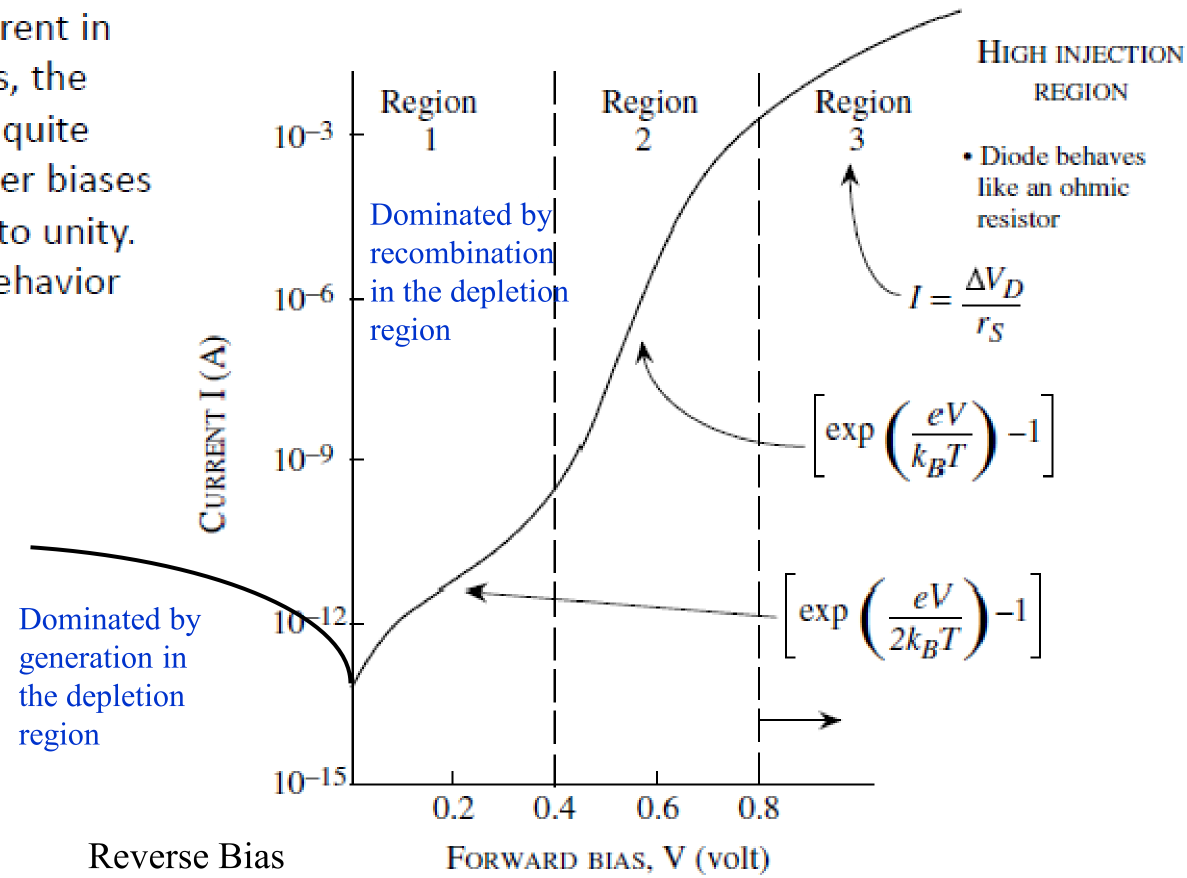
**P-N Junction Diode**

- Majority and minority carrier concentrations as a function of distance (x) at thermal equilibrium, forward bias and reverse bias (diagram and calculation)
- Majority and minority current distributions (as a function of x)
- Calculation of minority, majority and total currents
- Calculation of reverse situation current, and diode current from the reverse situation current
- Reverse breakdown mechanisms (Zener breakdown and Avalanche breakdown)
- Conditions required for Zener breakdown and Avalanche breakdown
- Breakdown voltage calculation

## P-N Junction Diode

- Diode dark current and effect of generation and recombination currents on the dark current

A log plot of the diode current in forward bias. At low biases, the recombination effects are quite pronounced, while at higher biases the slope becomes closer to unity. At still higher biases the behavior becomes more ohmic.



## P-N Junction Capacitance

- Calculation of junction capacitance (Cj) and storage capacitance (Cs) under different biases
- Dominated capacitance under different bias conditions
- Effect of capacitance on the device performance

**Metal-Semiconductor Contacts**

- Schottky contact (condition required for Schottky contact)
- Energy diagram before and after contact
- Calculation of depletion width in the semiconductor, Schottky barrier height and contact potential (energy barrier for carriers to inject from semiconductor to metal)
- Current calculation in Schottky diode
- Schottky diode vs p-n junction diode (advantages)
- Ohmic contact (condition required for ohmic contact)
- Energy diagram before and after contact
- Way to reduce barrier in ohmic contact

**BJT**

- Calculation of common-base current gain ($\alpha_0$), emitter efficiency ($\gamma$), base transport factor ($\alpha_T$), $I_{CBO}$
- Calculation of output current ($I_C$) from input current ($I_E$), gain ($\alpha$, $\beta$) and leakage current ($I_{CBO}$, $I_{CEO}$)
- Carrier distribution in emitter, base and collector region
- Calculation of each component of currents
- Methods to improve the BJT performance or emitter efficiency ($\gamma$)
- Three types of cut-off frequency (common base cut-off frequency, common emitter cut-off frequency and cut-off frequency)
- Calculation of transit time and cut-off frequency from transit time
- The methods to improve cut-off frequency

## MOS Capacitor

- Four modes of gate biasing (accumulation, flat band, depletion and inversion)
- Energy diagram and charge distribution of MOS structure at different gate biasing modes
- Calculation of semiconductor work function, metal work function, and $\Phi_{ms}$
- Calculation of $\Phi_B$, depletion width, band bending at the surface, threshold voltage
- Must clearly understand the concept of threshold voltage

## Other Tips

- Bring a calculator
- If you see the words such as “illustrate,” “sketch,” etc., you must draw a figure to explain.
- Don’t spend too much time on one question. There are a total of 4 questions, and you should not spend more than 30 min for each question.
- Constant values are given but you should already be familiar with these values. If you search for these values in the exam, you may not have enough time to answer all the questions.
- Show your calculation step-by-step. This will help you to get score even if your final answer is wrong.
- If you do not have enough time to do the numerical calculation, you can just show the step/approach with the corresponding formula. Only a small mark will be deducted for incompletion of the numerical calculation. However, you will lose a huge mark if you cannot finish the other questions.
- You can write down equations and formulas in cheating notes but should be familiar with equations beforehand and must know how and when to apply these equations.

$$n_0 = N_c \exp\left[-\frac{E_c - E_F}{k_B T}\right] \quad \Longleftrightarrow \quad E_c - E_F = -\frac{kT}{q}\ln\left(\frac{N_c}{n_0}\right)$$

$$n_0 = n_i \exp\left[\frac{E_F - E_i}{k_B T}\right] \quad \Longleftrightarrow \quad E_F - E_i = \frac{kT}{q}\ln\left(\frac{n_0}{n_i}\right)$$

$$p_0 = N_v \exp\left[-\frac{E_F - E_v}{k_B T}\right] \quad \Longleftrightarrow \quad E_F - E_V = -\frac{kT}{q}\ln\left(\frac{N_v}{n_0}\right)$$

$$p_0 = n_i \exp\left[\frac{E_i - E_F}{k_B T}\right] \quad \Longleftrightarrow \quad E_i - E_F = \frac{kT}{q}\ln\left(\frac{p_0}{n_i}\right)$$